

Pressure tells you where the lesion is: segmental tone heterogeneity and the penile deformities of hard flaccid syndrome

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Ninth in a series; see companion papers on the neural, effector and sensory conjectures.

Abstract

Background. Changes in penile shape or size were the single most commonly reported complaint in the largest hard flaccid syndrome (HFS) survey, at 92.3%. Hourglassing, bottlenecking and lateral tilt are described, and patients report that these resolve at full erection. No account of cavernosal tone explains a *regional* abnormality.

The deduction, which is the best in the series. A fibrous tunical plaque produces indentation that *worsens* with erection, because inelastic tissue cannot expand while the surrounding tunica does. A deformity that disappears at full erection must therefore be something maximal intracavernosal pressure can overcome. That is a clean, checkable inference from pressure-dependence to lesion class, and it is the sharpest reasoning anywhere in the conjecture set.

But it is too strong as stated. Pressure-dependence excludes an *inelastic* lesion. It does not exclude a structural one. Localised tunical thinning, an early compliant plaque, a fascial band within Buck's or dartos fascia, and segmental oedema are all structural and all overcome by pressure. The deduction should be narrowed to "not an inelastic tunical plaque", which is a considerably weaker conclusion than "not tunical".

And the syncytial argument runs backwards. The conjecture invokes connexin43 gap junction coupling as the substrate that turns uneven input into a coherent band. Coupling does the opposite: a well-coupled syncytium equalises tone across itself. Junctional conductance in human corporal smooth muscle is regulated — increased by protein kinase C activation and decreased by cyclic AMP — so $\alpha 1$ -adrenergic signalling would be expected to *strengthen* coupling and homogenise tone. Segmental heterogeneity requires **heterogeneous** coupling: fragmentation of the syncytium into semi-independent domains. That is a different and more specific claim.

The differential that must be excluded first. Partial segmental thrombosis of the corpus cavernosum produces exactly this presentation — a firm proximal segment with a flaccid distal penis in a young man, often after sexual or exercise microtrauma, frequently with a demarcating fibrous membrane. It is rare, has a specific magnetic resonance appearance, and requires entirely different management. A scoping review has already raised its possible association with HFS.

The missing denominator. No normative data exist on flaccid or partially erect penile contour in healthy men. An irregular flaccid contour observed by a hypervigilant self-observer may be unremarkable. We propose that study as the prerequisite; it is non-invasive and nobody has done it.

Keywords: hard flaccid syndrome; hourglass deformity; connexin43; gap junctions; syncytium; Peyronie's disease; partial segmental thrombosis of the corpus cavernosum.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. This paper discusses a differential diagnosis — partial segmental thrombosis of the corpus cavernosum — that has specific and different management; nothing here should be used for self-diagnosis, and anyone with a firm penile segment, penile or perineal pain of sudden onset should be assessed by a clinician rather than reasoning from this article.

1. Introduction

The most commonly reported complaint in the largest survey of hard flaccid syndrome (HFS) was not the hard flaccid state itself. Changes in penis shape or size were reported by 92.3% of 143 respondents, marginally ahead of a rigid penis when not erect at 90.9% [B] [1]. Patients describe hourglassing, bottlenecking and lateral tilt, and — critically — report that these deformities are present when flaccid or partially erect and disappear at full erection [E] [3].

Every other conjecture in this series concerns the tone of the corpus cavernosum as a whole. None of them explains a deformity, because a global change in tone should change the organ globally. A regional abnormality requires a regional explanation.

The conjecture examined here supplies one: that these deformities are produced by regional differences in smooth muscle tone within a gap-junction-coupled syncytium, and are not tunical lesions [3]. It arrives at that conclusion by a route that we regard as the best piece of reasoning in the whole set — an inference from the *pressure-dependence* of a deformity to the class of lesion producing it. This paper examines that inference, finds it sound but overstated, corrects the cell-biological argument that accompanies it, and identifies the differential diagnosis and the missing control data that stand between the conjecture and a test.

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *connexin43 corpus cavernosum*, *gap junction syncytium penile*, *hourglass deformity penis*, *Peyronie indentation erection*, *partial segmental thrombosis corpus cavernosum*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic. We searched specifically for structural causes of segmental penile firmness, and Section 6 reports what that returned.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence, clinical guideline, or uncontested descriptive human anatomy.
[B]	Single human study, human tissue study, case series, or human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic or logical inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. The deduction, and how far it actually reaches

3.1 The argument as made

In Peyronie’s disease a fibrous tunical plaque produces indentation or curvature that worsens with erection, because inelastic tissue cannot expand while the surrounding tunica does [A] [VERIFY] [9]. The deformity reported in HFS behaves oppositely: present when flaccid or partially erect, absent at full erection [E] [3]. Whatever produces it must therefore be something maximal intracavernosal pressure can overcome; and a tunical plaque is by definition not overcome by pressure. So the deformity is not a tunical plaque, and it is pressure-dependent [D].

We think this reasoning is correct and that it is the most valuable single move in the conjecture set, because it converts a subjective observation into a constraint on lesion class using nothing but mechanics. It deserves to be stated more carefully than it usually is, and the careful statement is weaker.

3.2 Correction: pressure-dependence excludes inelasticity, not structure

The step that does not hold is from “overcome by pressure” to “not tunical” — and more broadly, to “not structural”. What pressure-dependence excludes is an *inelastic* lesion: one stiffer than the surrounding tunica, which therefore constrains expansion more as pressure rises. It does not exclude structural abnormalities that are equally or *more* compliant than their surroundings, all of which would be effaced by high pressure [D]:

- **Localised tunical thinning or attenuation**, which would bulge rather than constrain at high pressure and might appear as a relative waist at low pressure.
- **An early, cellular, pre-fibrotic plaque**, which need not yet be inelastic.
- **A constricting band in a superficial layer** — Buck’s fascia, dartos, or a fibrosed superficial vein — external to the tunica and therefore stretched flat by tunical expansion.

- **Segmental oedema or lymphatic change**, which is displaced by pressure.
- **Regional differences in trabecular smooth muscle content** rather than tone — a fibrotically remodelled segment with less distensible sinusoidal tissue at low volume, overwhelmed at maximal filling.

The last of these is the most awkward, because it is a structural lesion that mimics a tone lesion at every pressure the conjecture's test would examine. **The corrected deduction is therefore: the deformity is not an inelastic tunical plaque. It may still be structural, and distinguishing tone from compliant structure requires imaging of the tissue, not only of the contour.**

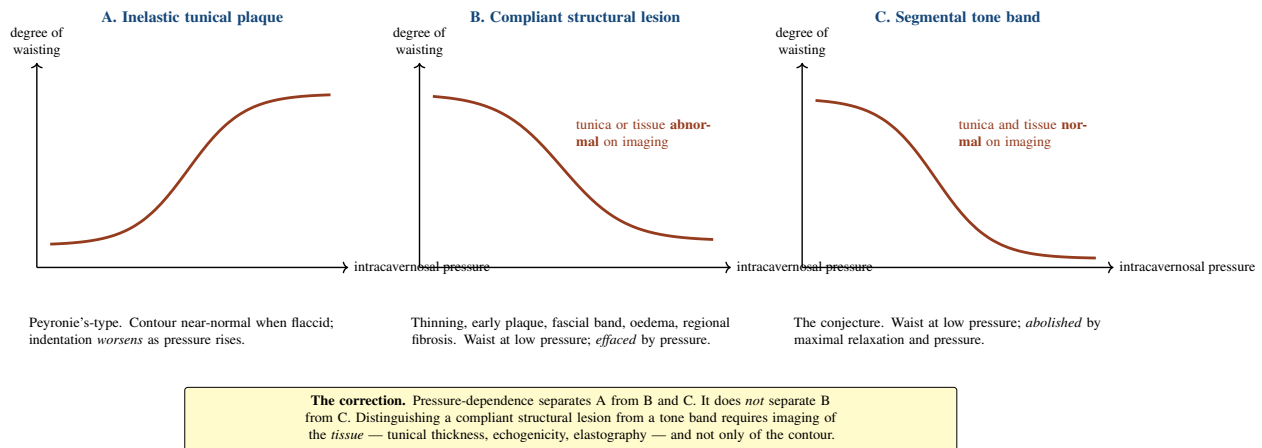


Figure 1. Contour behaviour of three lesion classes as intracavernosal pressure rises. These are hypothetical schematics illustrating a mechanical argument; they are not data, and no pressure–contour measurements exist for hard flaccid syndrome. An inelastic tunical plaque constrains expansion more as pressure rises, so indentation worsens (A). Any lesion equally or more compliant than its surroundings is effaced by pressure (B), as is a regional difference in smooth muscle tone (C). The conjecture's deduction correctly excludes A. It does not exclude B, which is the correction developed in Section 3.2.

4. The syncytial argument runs backwards

The conjecture supports its tone-band proposal by appeal to the coupled syncytium: corporal smooth muscle is sparsely innervated and its cells are joined by connexin43 gap junctions, and “a syncytium that behaves as coupled regional units is one in which uneven input produces a coherent band rather than scattered individual cell contraction” [3].

The underlying anatomy is correct and well established. Despite sparse autonomic innervation, corporal smooth muscle cells contract and relax synchronously; they express connexin43 and are joined by gap junctions in situ and in culture; and this coupling is proposed to be the mechanism of intercellular communication coordinating tissue contraction and relaxation [B] [4]. The cells are dye- and electrically coupled, and disruption of coupling with heptanol alters $\alpha 1$ -adrenergic contractile responses in human corporal tissue — a functional demonstration that intercellular communication matters for response generation [B] [VERIFY] [5, 6].

But coupling is what abolishes regional differences, not what creates them. That is what a syncytium is for. If every cell in the corpus cavernosum were strongly coupled to every neighbour, a focal excess of adrenergic input would be shared across the tissue and expressed as a modest global rise in tone — not as a band. The conjecture's own framing states the mechanism as coordination “across the tissue”, which is precisely the property that would prevent segmentation [D].

The regulation of coupling makes the tension sharper. Junctional conductance between human corporal smooth muscle cells is moderately voltage-sensitive, abolished reversibly by halothane, *increased* by the protein kinase C activator TPA, and *decreased* by isoproterenol [B] [5]. Since $\alpha 1$ -adrenergic signalling activates protein kinase C, sustained sympathetic overactivity would be expected to *strengthen* coupling and homogenise tone still further [D].

4.1 The repair

Segmental heterogeneity in a syncytium requires **heterogeneous coupling**: strong within a domain and weak at its boundaries, so that the tissue behaves as a set of semi-independent territories rather than as one sheet. On that account the lesion is not simply uneven input but *regional uncoupling*, and the boundaries of the deformity are the boundaries of the electrically isolated domain.

This is a more specific claim than the conjecture makes, and it has consequences [D]:

1. The waist should be **reproducible in location** within a patient, because it is set by a structural feature of the coupling

network rather than by transient input.

2. Its boundaries should be **sharp** rather than graded, since a coupling discontinuity produces a step.
3. Agents that reduce junctional conductance — including cyclic-AMP-elevating agents [5] — would be predicted to *accentuate* segmentation, whereas agents that increase coupling would smooth it. Intracavernosal prostaglandin E1 raises cyclic AMP, so this is in principle observable during a routine pharmacological erection test.

The third point is a genuine and, so far as we can find, unremarked prediction: **if segmentation is a coupling phenomenon, the waist should be most conspicuous during the early, partially tumescent phase of a cyclic-AMP-mediated pharmacological erection, and should efface as pressure rises**. It is also confounded, because pressure and cyclic AMP rise together in that setting, and disentangling them would require a comparison against a saline-infusion artificial erection at matched pressures [DATA NEEDED].

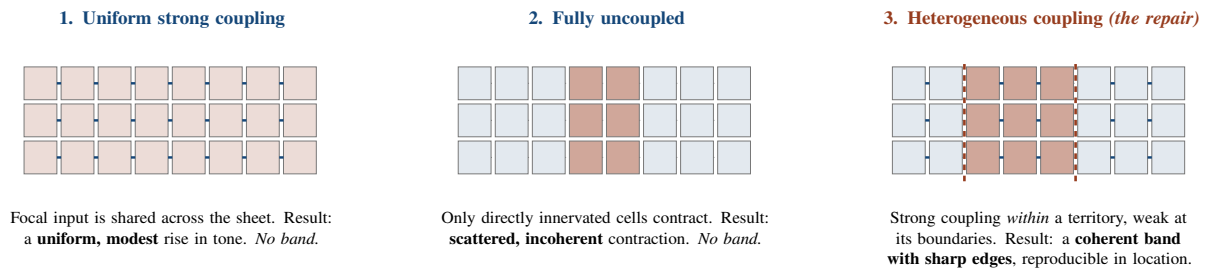


Figure 2. Why coupling alone cannot produce a segmental band. A well-coupled syncytium equalises tone across itself (panel 1); a fully uncoupled tissue produces scattered contraction confined to directly innervated cells (panel 2). A coherent band with defined edges requires coupling that is strong within a territory and weak at its boundaries (panel 3). The conjecture cites connexin43 coupling as its substrate but needs the discontinuity, not the coupling. Shading indicates tone; heavy links indicate functional gap junctions, dotted links indicate uncoupling. Diagram is schematic and illustrates an argument, not observed data.

5. The hypothesis

Formal statement, with the corrections applied

H9 (segmental tone heterogeneity). The hourglassing, bottlenecking and lateral tilt reported in hard flaccid syndrome are produced by regional differences in cavernosal smooth muscle tone, sustained by heterogeneous gap-junctional coupling that partitions the corporal syncytium into semi-independent territories. They are not inelastic tunical lesions, and they are effaced when maximal relaxation and maximal intracavernosal pressure override the regional difference.

H9-tone (the claim). The affected segment differs in tone but not in structure. Prediction: at graded pressures, cross-sectional area at the waist increases and normalises, while tunical thickness, echogenicity and tissue elasticity are indistinguishable from adjacent segments.

H9-compliant-structure (the principal competitor, not excluded by the conjecture's reasoning). The segment differs in structure but is compliant enough to be effaced by pressure. Prediction: identical contour behaviour, but detectable tissue abnormality on elastography or high-resolution imaging.

H9-null. The contour is within normal variation for a flaccid or partially erect penis, and is salient because it is being closely watched.

Auxiliary assumptions:

- **A1 (the observation is real).** The deformity exists, is reproducible, and resolves at full erection. Currently grade E self-report; not documented by imaging in any patient [DATA NEEDED].
- **A2 (it is abnormal).** The contour differs from that of healthy men at comparable tumescence. **No normative data exist** (Section 7).
- **A3 (coupling heterogeneity).** The corporal syncytium can partition into territories with sharp boundaries. Not demonstrated in any species; regional variation in connexin43 expression along the human corpus cavernosum has not been mapped [DATA NEEDED].

6. The differential that must be excluded first

One recognised condition produces almost exactly the presentation the conjecture attributes to tone heterogeneity, and it is not mentioned in the source article’s account of the deformities.

Partial segmental thrombosis of the corpus cavernosum (also termed partial priapism or partial segmental priapism) is a rare condition of young men — mean age around 32 in one summary — affecting the proximal and perineal portion of the corpus cavernosum, so that the affected segment is firm while the distal penis remains flaccid [B] [VERIFY] [10, 11]. In most reported cases a **well-defined membrane separates the proximal firm part from the distal flaccid part** [B] [VERIFY] [12]. Reported risk factors include microtrauma from exercise such as cycling, or **during sexual activity** [B] [VERIFY] [10]. It is usually unilateral. Fewer than about 100 cases have been reported [B] [VERIFY] [10].

A scoping review has examined the possible association between partial thrombosis of the corpus cavernosum, partial priapism and hard flaccid syndrome [B] [VERIFY: WE DID NOT RETRIEVE THIS REVIEW AND CANNOT REPORT ITS CONCLUSIONS] [14].

The overlap with the HFS presentation is uncomfortable: a young man, sexual or exercise microtrauma, penile or perineal pain, a segmentally firm penis with a demarcation, and a distal segment that behaves differently from the proximal one. The distinguishing features are that partial segmental thrombosis is unilateral and proximal, has a characteristic magnetic resonance appearance, and **requires different management, typically anticoagulation** [B] [VERIFY] [11, 13].

We are not proposing that HFS is partial segmental thrombosis; it is far too rare and the imaging in HFS is described as normal [2]. We are proposing that **no study of segmental penile deformity in this population is interpretable without magnetic resonance imaging to exclude it**, and that a conjecture about segmental firmness which does not mention the one established cause of segmental firmness is incomplete.

Table 2. Differential diagnosis of segmental penile firmness or waisting, with distinguishing features.

Entity	Behaviour with erection	Imaging	Distinguishing features
Peyronie’s disease	Indentation or curvature worsens	Tunical plaque, often calcified	Palpable plaque; pain on erection early; deformity absent or minimal when flaccid
Partial segmental thrombosis of the corpus cavernosum	Segment persistently firm; demarcating membrane	Characteristic MRI: thrombus, absent flow in a proximal segment	Unilateral, proximal/perineal, acute pain onset, young man, post-microtrauma; different management
Localised tunical thinning or early compliant plaque	Waist effaced by pressure	Tunical thickness abnormality on high-resolution ultrasound	Structural but compliant; not excluded by pressure-dependence
Regional fibrotic remodelling of trabecular tissue	Waist effaced at maximal filling	Altered echogenicity; elastography abnormality	Mimics a tone band at every pressure the contour test examines
Superficial fascial or venous band	Waist effaced by tunical expansion	External to tunica on ultrasound	Palpable superficially; mobile relative to tunica
Segmental tone heterogeneity (H9)	Waist abolished by maximal relaxation and pressure	Tunica and tissue normal	Reproducible location; sharp boundaries; normal tissue on all modalities
Normal variation	Variable	Normal	No normative data exist (Section 7)

7. The missing denominator

We could identify no normative dataset describing the contour of the flaccid or partially erect penis in healthy men: no measurements of circumference variation along the shaft, no description of how common visible waisting is, and no characterisation of how contour changes across the tumescence range [DATA NEEDED].

This is a serious problem for H9 and for the reported symptom itself. The flaccid penis is a low-pressure, partly collapsed structure whose surface contour reflects skin, fascia, superficial veins and variable trabecular filling; some irregularity is unremarkable. The observation is being made by men who are, by the nature of the condition, examining their genitals frequently and anxiously — exactly the circumstance in which normal variation becomes salient and is recorded as new [D].

The normative study is non-invasive, requires no drug or procedure, could be completed in a season, and would inform the interpretation of a symptom reported by 92.3% of patients [1]. We regard it as the prerequisite for everything else in this paper, and note that it would be useful regardless of whether H9 is true.

8. Predictions and tests

8.1 Prerequisite: normative contour data

Prediction P0. A defined threshold of shaft circumference variation, established in healthy men across the tumescence range, will distinguish men with HFS from controls.

Method: standardised photogrammetry or ultrasound measurement of circumference at fixed intervals along the shaft, in healthy volunteers, at flaccid, partially tumescent and fully erect states, with age stratification. Tumescence can be graded non-invasively with a vacuum device or by self-stimulation to a defined erection hardness score, avoiding intracavernosal injection in healthy controls.

8.2 Primary test: dynamic ultrasound at graded pressures

Prediction P1. In men with HFS, the site of narrowing is reproducible within a patient across sessions; its cross-sectional area increases progressively with intracavernosal pressure and normalises at full erection; and the tunica albuginea remains sonographically normal at the waist.

This is the conjecture’s own proposed test [3] and it is among the most feasible in the whole series, because penile duplex Doppler ultrasound with pharmacologically induced erection is a routine urological examination. Three additions are needed to make it interpretable:

- **Tissue imaging, not only contour.** Shear-wave elastography or high-resolution assessment of trabecular echogenicity at the waist versus adjacent segments, because contour behaviour alone cannot separate a tone band from a compliant structural lesion (Section 3.2, Figure 1).
- **Magnetic resonance imaging** in any patient with a firm proximal segment, to exclude partial segmental thrombosis (Section 6).
- **Blinded measurement** against age-matched controls examined identically, since the outcome is a contour judgement.

Prediction P2. The waist corresponds to a nerve distribution territory or to a remembered injury site. This is the conjecture’s suggestion [3] and it is worth recording, but we note that the segmental innervation territories of the human corpus cavernosum have not been mapped in a way that would allow the correspondence to be judged [DATA NEEDED].

8.3 The coupling prediction

Prediction P3. Segmentation is most conspicuous during early tumescence in a cyclic-AMP-mediated pharmacological erection and effaces as pressure rises, and is less conspicuous at matched pressures during a saline-infusion artificial erection.

This follows from the observation that junctional conductance is decreased by cyclic-AMP-elevating agents [5] and would, if confirmed, provide evidence that the segmentation is a coupling phenomenon rather than simply a tone difference. It requires cavernosometry with saline infusion, which is invasive and now infrequently performed, and we regard it as a second-line study.

8.4 The lateral tilt prediction

Prediction P4. Lateral tilt reflects asymmetric tone between the paired corpora, and should therefore vary with overall tone — being most marked when the hard flaccid state is most marked, and absent at full erection.

We note the base-rate problem again: some degree of lateral deviation of the erect penis is common and frequently congenital, and a study would need to establish whether the tilt is new, whether it is state-dependent, and whether it exceeds population variation [DATA NEEDED].

Table 3. Predictions of H9 compared with the principal alternatives.

Observation	H9 (tone band)	Compliant structural lesion	Inelastic tunical plaque	Discriminating value
Waist effaced as pressure rises	Predicted	Predicted	Refuted	High vs. plaque; none vs. compliant structure
Tunica sonographically normal at the waist	Predicted	Not predicted	Not predicted	Moderate-high
Normal elastography at the waist	Predicted	Not predicted	Not predicted	High — the test that separates H9 from its competitor
Waist reproducible in location across sessions	Predicted (coupling form)	Predicted	Predicted	None
Sharp rather than graded boundaries	Predicted (coupling form)	Predicted	Predicted	Low
Segmentation accentuated in early cAMP-mediated tumescence	Predicted	Not predicted	Not predicted	Moderate; confounded by pressure
Contour within normative range for healthy men	Refutes	Refutes	Refutes	Decisive, and untested

9. Alternative explanations

1. **Normal variation observed under scrutiny.** The most likely explanation of a subtle, pressure-effaced contour irregularity in a young man with no imaging abnormality, and the one that no existing data can exclude. We regard it as the leading hypothesis until normative data exist.
2. **A compliant structural lesion.** Not excluded by the conjecture's reasoning, and mimicking it on every contour-based measure (Section 3.2).
3. **Partial segmental thrombosis.** Rare but recognised, produces segmental firmness in exactly this population, and requires different management (Section 6).
4. **Regional fibrotic remodelling.** If chronic hypoxia drives collagen deposition unevenly [3], a segment with reduced smooth muscle and increased collagen would be less distensible at low volume and could be overwhelmed at maximal filling — indistinguishable from a tone band by contour alone, distinguishable by elastography.
5. **Volume rather than tone.** Segmental differences in sinusoidal filling, from local venous or lymphatic factors, would produce contour change without any difference in smooth muscle tone.
6. **The observation may not survive measurement.** “Resolves at full erection” has never been documented by imaging in any patient. It is possible that on formal examination the waist does not resolve, does not appear, or is not where the patient reports it.

10. Implications

No clinical implications are drawn, except one point of safety: segmental penile firmness of acute onset in a young man after sexual activity warrants clinical assessment, because partial segmental thrombosis has specific management and this article is not a basis for reasoning about an individual case.

Three research implications follow.

First, get the denominator. Normative flaccid and partially erect penile contour data would inform the interpretation of the most commonly reported symptom in this condition and do not currently exist. This is the cheapest high-value study in the entire series.

Second, image the tissue, not only the shape. The conjecture's proposed examination cannot distinguish its own hypothesis from a compliant structural lesion. Adding elastography converts it into a discriminating test.

Third, exclude the thrombosis. Any imaging study of segmental deformity in this population should include magnetic resonance imaging.

11. Limitations

1. **The observation is grade E.** The pressure-dependence of the deformity — the premise on which the entire deduction rests — is patient self-report and has never been documented by imaging.
2. **No denominator.** There are no normative data on penile contour in healthy men at any degree of tumescence.
3. **The deduction is narrower than stated.** It excludes an inelastic tunical plaque, not structural abnormality in general.
4. **The syncytial argument requires the opposite of what is cited.** Coupling homogenises; segmentation requires regional uncoupling, which has not been demonstrated in corporal tissue in any species.
5. **Regional connexin43 distribution along the human corpus cavernosum has not been mapped,** so the domain structure the corrected hypothesis requires is entirely conjectural.
6. **The innervation territories are unmapped,** so the conjecture's prediction that the waist corresponds to a nerve territory cannot currently be evaluated.
7. **The differential was incomplete.** A recognised cause of segmental penile firmness in this exact population was not considered by the conjecture.
8. **Several citations are flagged unverified,** particularly those concerning partial segmental thrombosis, which were traced through case reports and a review we did not retrieve in full.
9. **The clinical significance is unclear.** A deformity that resolves at full erection may be distressing but is of uncertain functional consequence, and this should temper the priority assigned to investigating it.

10. **Confirmation risk.** This paper was written to develop a specific conjecture, and it credits that conjecture's central deduction as the best reasoning in the series before narrowing it substantially. Readers should check the mechanical argument in Section 3.2 for themselves rather than accepting either the praise or the correction on our authority.

12. Refutation criteria

- **Refuted.** Measured shaft contour in men with HFS falls within the normative range for healthy men at matched tumescence.
- **Refuted.** Dynamic imaging shows the waist *worsening* with rising intracavernosal pressure, indicating an inelastic lesion.
- **Substantially refuted.** Elastography or high-resolution imaging shows abnormal tissue at the waist, indicating a compliant structural lesion rather than a tone band.
- **Substantially weakened.** The reported waist is not reproducible in location across sessions, or is not present on formal examination.
- **Weakened.** Segmental firmness in a given patient proves to be partial segmental thrombosis on magnetic resonance imaging.
- **Weakened (coupling form).** Connexin43 expression is shown to be uniform along the human corpus cavernosum with no capacity for domain structure.

13. Conclusion

This conjecture contains the best single piece of reasoning in the set. It takes a subjective observation — the deformity goes away at full erection — and converts it into a constraint on what kind of lesion can be responsible, using only mechanics. That is the right way to reason in a field with almost no data, and it deserves to be imitated.

It also reaches further than the mechanics allow. Pressure-dependence excludes a lesion *stiffer* than its surroundings. It says nothing about a lesion equally or more compliant than its surroundings, and localised tunical thinning, an early cellular plaque, a superficial fascial band, segmental oedema and regional fibrotic remodelling are all in that category. The corrected conclusion — not an inelastic tunical plaque — still excludes Peyronie's disease, which is worth something, but it leaves the conjecture's principal competitor untouched. Adding tissue imaging to the proposed examination fixes this, and turns a test that could only have confirmed the hypothesis into one that could discriminate it.

The cell biology needs inverting. Gap junction coupling is cited as the substrate that turns uneven input into a coherent band, but coupling is what *prevents* regional differences; that is what a syncytium does. Junctional conductance in human corporal smooth muscle is increased by protein kinase C activation, which α 1-adrenergic signalling recruits, so sympathetic overactivity should homogenise rather than segment. What a band requires is a discontinuity: strong coupling within a territory, weak at its edges. That is a sharper hypothesis, it predicts reproducible location and sharp boundaries, and it generates an unexpected prediction about cyclic-AMP-mediated erections that nobody has looked for.

Two things stand between all of this and an answer, and neither is difficult. Nobody has measured the contour of a normal flaccid penis, so nobody knows whether the observation describes a finding or a normal variant noticed under scrutiny. And a recognised condition — partial segmental thrombosis of the corpus cavernosum — produces segmental firmness in young men after sexual microtrauma, has a specific magnetic resonance appearance, and needs anticoagulation rather than a theory. The normative study costs almost nothing. The imaging is routine. Until both are done, the deformity that 92.3% of these patients report remains the most frequently described and least investigated feature of the condition.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** The partial segmental thrombosis literature cited here was assembled from case reports and should be replaced with a systematic account before submission, given its clinical implications.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [3]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the

argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. **Segmental firmness of the penis, particularly of acute onset and with pain, requires clinical assessment; partial segmental thrombosis of the corpus cavernosum is a recognised condition with specific management and should not be self-diagnosed or self-excluded on the basis of this article.** Penile duplex ultrasound with pharmacologically induced erection and cavernosometry are invasive procedures appropriate only under qualified medical supervision.

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